

# Integrating UX and UI Design Practices into Software Development

## Introduction: The Importance of User-Centric Design in Software Development

The landscape of software development has undergone a significant transformation over the years. Initially, the primary focus was on technical feasibility and functionality. However, with the increasing sophistication of technology and the proliferation of software products, the emphasis has shifted towards creating user-centric experiences. Today, the success of a software product is not solely determined by its technical capabilities but also by its ability to meet the needs and expectations of its users. Users expect products that are not only functional but also intuitive, enjoyable, and accessible.<sup>1</sup>

Addressing these user expectations requires a deep understanding of their needs, behaviors, and motivations. This is where the disciplines of User Experience (UX) and User Interface (UI) design come into play. UX design encompasses the overall interaction a user has with a company, its services, and its products, focusing on making these interactions effective and enjoyable.<sup>6</sup> UI design, on the other hand, centers on the visual and interactive elements of a product, aiming to create interfaces that are aesthetically pleasing and easy to use.<sup>8</sup>

This report aims to provide a comprehensive understanding of UX and UI design, detailing their respective roles within the software development pipeline. Specifically, it will address how these practices can be integrated into the user's company's existing pipeline, which currently includes requirements, development, testing, and deployment phases. The report will also explore the impact of this integration on the responsibilities of the Delivery and Development teams, drawing upon research and established best practices in the field.

## Understanding User Experience (UX) Design

### Defining UX: Core Principles and Objectives

User Experience (UX) design is a multifaceted discipline focused on the holistic interaction between an end-user and a company, its services, and its products.<sup>6</sup> It encompasses all aspects of the user's perception and response resulting from the use or anticipated use of a system, product, or service.<sup>11</sup> The goal of UX design is to optimize the user's journey, ensuring it is intuitive, easy, and enjoyable.<sup>8</sup>

Several core principles underpin effective UX design. User-centricity is paramount, emphasizing the need to create products and services that solve user problems by putting the user's needs first.<sup>24</sup> Usability, which measures the ease with which a user can carry out a task within a product or service, is another critical principle.<sup>24</sup> Accessibility ensures that users of a

full range of abilities can access and use products.<sup>15</sup> Consistency in design helps eliminate confusion and makes products more intuitive and learnable.<sup>24</sup> Hierarchy, both in information architecture and visual presentation, shapes how users navigate a product.<sup>25</sup> Understanding the context in which users interact with a product is also essential for providing a relevant experience.<sup>25</sup> Finally, user control empowers users by giving them the ability to manage their interactions and recover from errors.<sup>25</sup>

The objectives of UX design are centered around meeting the exact needs of the customer without fuss or bother, leading to products that are a joy to own and use.<sup>9</sup> This involves enhancing user satisfaction and loyalty by providing utility, ease of use, and pleasure in the interaction.<sup>2</sup> Ultimately, UX design aims to create a positive response, encouraging users to stay longer and return often by making the product visually appealing, engaging, and simple, with effortless navigation.<sup>1</sup>

The broad scope of UX, as highlighted by definitions from the Nielsen Norman Group and Interaction Design Foundation, extends beyond the digital interface to encompass the entire user journey and their overall interaction with the company.<sup>6</sup> This suggests that by incorporating UX considerations early in the development process, the company can gain a more comprehensive understanding of user needs that extends beyond just the software's features, potentially influencing aspects like documentation and support. Furthermore, the core principles of UX emphasize a balance between the functional aspects of a product, such as usability and usefulness, and the emotional aspects, including desirability and satisfaction.<sup>1</sup> This indicates that integrating UX into the requirements phase should focus not only on what the software will do but also on how it will feel for the user to interact with it.

## **The Role of UX in the Software Development Lifecycle: Focus on the Requirements Phase**

UX design plays a crucial role throughout the software development lifecycle (SDLC), ensuring that the final product meets user needs and provides a positive experience.<sup>32</sup> Its impact is particularly significant during the requirements gathering and analysis phase, where understanding user needs and preferences is paramount.<sup>33</sup>

User research is the first and arguably the most critical step in incorporating UX into software development.<sup>33</sup> It involves various methodologies to understand the user's needs, preferences, behaviors, motivations, and pain points.<sup>33</sup> By gathering both qualitative and quantitative data about the target audience, UX designers can create user-centered designs that align with user expectations.<sup>37</sup> This research helps in defining the software's objectives and requirements, ensuring that the product aligns with user expectations and effectively solves their problems.<sup>33</sup>

UX practices also inform the definition of software requirements through the use of user stories.<sup>53</sup> User stories are short, plain-language descriptions of how a software feature should work from the end user's perspective, detailing what they want to achieve and what they expect from the product.<sup>53</sup> This user-centric approach ensures that the development team

focuses on building features that provide real value to users.<sup>53</sup>

Information architecture (IA) plays a vital role in the requirements phase by focusing on how information is organized, structured, and presented to users.<sup>37</sup> A well-defined IA ensures that content is findable and navigable, allowing users to quickly locate the information they need.<sup>15</sup>

Low-fidelity prototyping is another key UX practice applied during the requirements phase.<sup>33</sup>

These early-stage prototypes, which can range from hand-drawn sketches to basic digital wireframes, serve as an early representation of the software, allowing the team to test its usability and functionality before significant development effort is invested.<sup>33</sup> Feedback gathered from stakeholders and potential users on these prototypes helps identify issues with the software's design, navigation, or functionality early in the process.<sup>33</sup>

Integrating UX in the requirements phase can substantially decrease the likelihood of developing the wrong product or features, thereby conserving time and resources in subsequent stages.<sup>42</sup> User research provides invaluable insights into the actual needs of users, which may often differ from the assumptions held by internal stakeholders.<sup>45</sup>

Furthermore, the utilization of user stories and low-fidelity prototypes during this phase fosters enhanced communication and collaboration among business analysts, stakeholders, and the development team.<sup>33</sup> User stories provide a user-centered lens through which to view requirements<sup>53</sup>, while prototypes transform abstract concepts into tangible visuals, facilitating early feedback and ensuring alignment across various teams.<sup>33</sup>

## Understanding User Interface (UI) Design

### Defining UI: Focus on Visual and Interactive Elements

User Interface (UI) design is the process designers use to build interfaces in software or computerized devices, with a focus on looks or style.<sup>8</sup> UI design centers on the visual and interactive elements of a product, aiming to create interfaces that users find easy to use and pleasurable.<sup>14</sup>

Aesthetics and visual appeal are critical in UI design, contributing significantly to a positive user experience.<sup>8</sup> The visual look and feel of the UI sets the tone for the entire user experience.<sup>12</sup> UI design also plays a crucial role in ensuring usability and ease of interaction with the product.<sup>12</sup> A well-designed UI guides the user through a product, making each interaction feel natural and logical.<sup>65</sup>

Interactive elements are another key focus of UI design, engaging users and facilitating task completion.<sup>8</sup> These elements, such as buttons, icons, and menus, allow users to interact with the software and achieve their goals.<sup>69</sup> Furthermore, UI design is instrumental in reinforcing brand identity and creating a consistent visual language across the product.<sup>8</sup> Consistency in UI elements, such as colors, typography, and branding, helps build trust and familiarity for the user.<sup>72</sup>

While UX design lays the groundwork for a positive user experience, UI design is critical for the product's initial impression and can significantly impact user engagement and trust.<sup>8</sup> A

visually appealing and well-organized interface can encourage users to explore and interact with the software. Moreover, consistency in UI elements and design patterns is vital for usability and learnability.<sup>14</sup> Users tend to prefer familiar interfaces, and inconsistencies can lead to frustration.<sup>24</sup>

## **The Role of UI Design in the Software Development Process: Integration during Development and Styling**

UI design plays a crucial role in the software development process, particularly during the development and styling stages.<sup>32</sup> This is where the visual and interactive elements defined in the UI design are brought to life.

High-fidelity prototypes are extensively used during this phase to visualize the final look and feel of the product and to test detailed interactions.<sup>33</sup> These prototypes go beyond the basic structure of low-fidelity prototypes, incorporating visual design elements like color palettes, typography, and imagery.<sup>41</sup> They also allow for the testing of specific user interactions, ensuring a smooth and intuitive flow.<sup>38</sup>

UI designers provide essential guidance on visual design elements, including the selection of color palettes, typography, imagery, and iconography.<sup>8</sup> They ensure that these elements not only are aesthetically pleasing but also align with the brand identity and enhance the overall user experience.<sup>62</sup>

Interaction design, a key component of UI, defines how users interact with the UI elements and ensures a seamless and intuitive flow.<sup>8</sup> This involves considering the behavior of UI elements in response to user actions, such as button clicks, form submissions, and navigation.<sup>8</sup>

Responsive design is also a critical aspect of UI during the development phase.<sup>8</sup> It ensures that the product adapts seamlessly to different screen sizes and devices, providing a consistent and optimal user experience across various platforms.<sup>12</sup>

Integrating UI design support during development ensures that the visual and interactive aspects are considered early and are feasible to implement.<sup>35</sup> Close collaboration between UI designers and developers is crucial for a successful outcome.<sup>33</sup> Early involvement of UI designers allows developers to understand the design intent and address any technical limitations proactively.<sup>38</sup> This collaboration ensures that the final product accurately reflects the design specifications.<sup>33</sup> Furthermore, high-fidelity prototypes serve as an essential communication tool among designers, developers, and stakeholders.<sup>33</sup> They provide a clear vision of the final product, facilitating effective feedback and iterative improvements.<sup>33</sup>

## **Impact on Team Responsibilities**

### **The Evolving Role of the Delivery Team with Integrated UX in Requirements**

A software Delivery Team is typically responsible for the entire process of delivering software

products from end to end, often comprising a diverse group of professionals with complementary skills.<sup>88</sup> These responsibilities can include understanding business needs, defining technical requirements, detailing structural and functional designs, creating and integrating code, testing quality and functionality, and releasing the product.<sup>88</sup>

Incorporating UX practices in the requirements phase will likely evolve these responsibilities, requiring a greater emphasis on understanding and advocating for the end-user from the very beginning of the project.<sup>88</sup> This will necessitate increased collaboration between UX designers (who may be new team members or existing Business Analysts with expanded roles), Product Owners, and other members of the Delivery Team during the initial stages of defining the project.<sup>88</sup>

The Delivery Team will likely adopt a more user-centered approach to defining project scope and priorities, leveraging insights gained from UX research.<sup>88</sup> This means that decisions about what features to include and how the product should function will be more heavily influenced by a deep understanding of user needs and pain points.

The focus of the Delivery Team will likely broaden to encompass a more holistic view of the product, considering not just technical feasibility but also user needs and satisfaction right from the outset.<sup>88</sup> This may require the team to undergo training on fundamental UX principles or to integrate dedicated UX roles within the team structure. Furthermore, Business Analysts within the Delivery Team may need to acquire skills in UX research methodologies and low-fidelity prototyping techniques to effectively contribute during the requirements phase, aligning with the plan to extend UX knowledge across Business Analytics.<sup>88</sup>

## **The Evolving Role of the Development Team with Integrated UI Design Support**

The standard responsibilities of a Development Team in software production typically involve translating requirements into functional code, testing that code to ensure quality, and maintaining and updating the software.<sup>88</sup>

The inclusion of UI design support during the development phase will significantly affect these responsibilities, fostering a closer working relationship between UI designers and developers.<sup>33</sup> Developers will rely on high-fidelity prototypes and UI style guides provided by designers for accurate implementation of the user interface.<sup>33</sup> This collaboration will likely lead to developers gaining a better understanding of the user interface and the rationale behind design decisions through more frequent and direct interaction with UI designers.<sup>35</sup>

The availability of detailed high-fidelity prototypes and comprehensive UI style guides should streamline the development process by providing clear and specific guidelines, thereby reducing ambiguity and the potential for rework related to the user interface.<sup>38</sup> This enhanced clarity should enable developers to implement the intended visual design and user interactions more efficiently.

## **Successful Integration: Case Studies and Best**

## Practices

While the provided material does not offer specific case studies of named companies detailing their UX/UI integration processes, it does highlight the importance of UI/UX service companies, implying that many organizations rely on these specialists to integrate design thinking into their development pipelines.<sup>104</sup> These companies often employ methodologies and tools that foster collaboration between design and development teams.

Agile methodologies, with their emphasis on iterative development and close collaboration, are frequently adopted to integrate UX and UI design effectively.<sup>21</sup> Design sprints, which are time-bound, focused efforts to solve specific design problems, can also be valuable in this integration.<sup>21</sup> The use of shared design systems and UI component libraries is another best practice that ensures consistency and efficiency across both design and development efforts.<sup>24</sup>

Companies that successfully integrate UX/UI often experience numerous benefits, including improved user satisfaction, increased efficiency in the development process, and potentially reduced development costs due to fewer errors and less rework.<sup>2</sup> Furthermore, the implementation of continuous integration and continuous delivery (CI/CD) pipelines can facilitate the smooth and frequent deployment of user-centric designs.<sup>113</sup>

The successful integration of UX and UI design often involves adopting agile methodologies, which allow for iterative design and development with continuous user feedback.<sup>50</sup> This iterative approach ensures that the product evolves based on user needs and feedback, leading to a more user-friendly and effective final product. Additionally, the use of shared design systems and UI component libraries can significantly enhance both the efficiency and consistency of the design and development processes.<sup>24</sup> These systems provide a centralized repository of reusable design elements and guidelines, ensuring a unified user experience across the product and reducing the time and effort required for both design and development tasks.

## UX vs. UI: Clarifying the Differences and Overlaps

While often used interchangeably, UX (User Experience) and UI (User Interface) design are distinct yet interconnected disciplines.<sup>8</sup> UX design focuses on the overall experience a user has with a product, ensuring it is intuitive and meets their needs, essentially focusing on how a product *feels*.<sup>116</sup> UI design, on the other hand, deals with the visual and interactive elements of a product, making it appealing and easy to navigate, primarily concerned with how a product *looks*.<sup>116</sup>

The design process for UX typically comes first, involving user research, information architecture, and wireframing to establish the product's structure and functionality.<sup>116</sup> UI design follows, adding the visual design and interactive elements to make the product polished and engaging.<sup>116</sup> Testing in UX focuses on functionality, intuitiveness, and problem-solving, while UI testing emphasizes visual appeal, branding consistency, and the ease of use of interactive elements.<sup>116</sup> Success in UX is measured by user satisfaction,

usability, and task completion, whereas UI success is gauged by visual appeal, ease of navigation, and the enhancement of the user experience.<sup>116</sup>

Despite their differences, UX and UI design share several overlaps.<sup>8</sup> Both require empathy for end-users and consideration of how users want to interact with a product.<sup>116</sup> Both UX and UI designers often work in cross-functional teams, collaborating with graphic designers and developers.<sup>116</sup> Skills such as prototyping and wireframing are common to both fields.<sup>118</sup>

Ultimately, a seamless and successful digital product requires UX and UI to work in harmony.<sup>116</sup>

An analogy often used to illustrate their relationship is that UX is like the architect who designs the blueprint of a house, ensuring it is functional and meets the needs of the occupants, while UI is like the interior designer who focuses on the aesthetics and the look and feel of the house.<sup>13</sup> Just as a well-designed house needs both a solid structure and appealing interiors, a successful software product requires both effective UX and engaging UI.

The user's plan to incorporate UX in the requirements phase and UI in the development phase aligns with the general understanding that UX often precedes UI in the design process, as UX provides the foundational understanding of user needs and the product's structure.<sup>8</sup> However, continuous collaboration between UX and UI designers throughout the entire development lifecycle is essential to ensure a cohesive and user-centered product.<sup>67</sup>

The following table summarizes the key differences and overlaps between UX and UI design:

| Feature      | UX Design                                                                   | UI Design                                                       | Overlap                                                             |
|--------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| Focus        | Overall user experience, usability, usefulness                              | Visual appearance, interactivity, aesthetics                    | User-centered design, understanding user needs                      |
| Process      | Research, analysis, information architecture, wireframing                   | Visual design, interaction design, prototyping, styling         | Iterative design, prototyping, collaboration with development teams |
| Deliverables | User personas, user flows, wireframes, low-fidelity prototypes              | Visual mockups, high-fidelity prototypes, style guides, UI kits | Prototypes, design specifications                                   |
| Skills       | User research, information architecture, usability testing, problem-solving | Visual design, typography, color theory, interaction design     | Empathy, communication, design thinking, attention to detail        |
| Goal         | Create effective, efficient, and satisfying user experiences                | Create visually appealing, intuitive, and usable interfaces     | Deliver a product that meets user needs and business objectives     |

## Conclusion and Recommendations

Integrating UX and UI design practices into the software development pipeline holds significant potential for enhancing the quality and success of the company's software products. By focusing on the user's needs and preferences throughout the development lifecycle, the company can create products that are not only functional but also enjoyable and easy to use, leading to increased user satisfaction and loyalty.

Based on the research and analysis, the following actionable recommendations are provided for the user's company:

1. **Establish Clear Definitions and Understanding:** Ensure that all team members within the organization have a clear and consistent understanding of what UX and UI design entail and the value they bring to the software development process. This can be achieved through workshops, training sessions, and the development of internal guidelines.
2. **Implement User Research in the Requirements Phase:** Introduce user research activities as a standard part of the requirements gathering process. This could involve conducting user interviews, surveys, usability testing of existing products, and analyzing user feedback to gain a deep understanding of user needs and pain points.
3. **Introduce Low-Fidelity Prototyping Early:** Incorporate the creation of low-fidelity prototypes during the requirements phase. These prototypes should be used to visualize initial concepts, test user flows, and gather early feedback from stakeholders and potential users.
4. **Provide UI Design Support and High-Fidelity Prototyping During Development:** Integrate UI designers into the development phase to provide ongoing support on visual design and interaction patterns. Utilize high-fidelity prototypes to communicate the final look and feel of the product and to test detailed interactions before final implementation.
5. **Foster Collaboration and Communication:** Establish clear channels for collaboration and communication between UX, UI, and development teams. Encourage regular meetings, shared documentation, and the use of collaborative design tools to ensure a seamless flow of information and feedback throughout the development process.
6. **Consider Dedicated UX/UI Roles or Training:** Evaluate the need for dedicated UX and UI designers within the organization. If hiring new roles is not immediately feasible, consider providing training opportunities for existing team members, particularly Business Analysts and developers, to enhance their UX and UI knowledge and skills.
7. **Establish a Design System or UI Component Library:** Develop a comprehensive design system or UI component library that includes reusable UI elements, design patterns, and style guidelines. This will help ensure consistency across the product and streamline both the design and development processes.
8. **Iterate Based on Feedback and Results:** Continuously monitor user feedback and analyze product usage data to identify areas for improvement. Adopt an iterative approach to both UX and UI design, making adjustments based on real user data to enhance the overall user experience.

By implementing these recommendations, the user's company can successfully integrate UX and UI design practices into its software development pipeline, ultimately creating more

user-centric, successful, and competitive software products.

## Cytowane prace

1. www.walkme.com, otwierano: kwiecień 11, 2025, <https://www.walkme.com/blog/ux-5-goals-set/#:~:text=UX%20design%20is%20a%20about,its%20features%20should%20feel%20effortless.>
2. What is UX design? 15 user experience design experts weigh in - UserTesting, otwierano: kwiecień 11, 2025, <https://www.usertesting.com/blog/what-is-ux-design-15-user-experience-experts-weigh-in>
3. UI/UX Design Goals: Crafting Digital Experiences - axon.dev, otwierano: kwiecień 11, 2025, <https://www.axon.dev/blog/ui-ux-design-primary-goals-and-approaches->
4. 5 User Experience Goals for UX Designers - WalkMe, otwierano: kwiecień 11, 2025, <https://www.walkme.com/blog/ux-5-goals-set/>
5. The Primary Goals of Good UX Design - Avallo Web Development, otwierano: kwiecień 11, 2025, <https://www.avallo.com/blog/the-primary-goals-of-good-ux-design/>
6. in.nau.edu, otwierano: kwiecień 11, 2025, <https://in.nau.edu/experience-design/what-is-user-experience/#:~:text=%E2%80%9DUser%20experience%E2%80%9D%20encompasses%20all%20aspects.to%20form%20their%20user%20experience.>
7. What is User Experience? | NAU Experience Design, otwierano: kwiecień 11, 2025, <https://in.nau.edu/experience-design/what-is-user-experience/>
8. UX vs. UI Design: What's the Difference? [2025 Guide] - CareerFoundry, otwierano: kwiecień 11, 2025, <https://careerfoundry.com/en/blog/ux-design/the-difference-between-ux-and-ui-design-a-laymans-guide/>
9. What Is User Experience (UX) Design - Coco Solution, otwierano: kwiecień 11, 2025, <https://cocosolution.com/en/blog/user-experience-ux/>
10. What is User Experience (UX) Design? — updated 2025 | IxDF, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/topics/ux-design>
11. User experience - Wikipedia, otwierano: kwiecień 11, 2025, [https://en.wikipedia.org/wiki/User\\_experience](https://en.wikipedia.org/wiki/User_experience)
12. User interface design - Wikipedia, otwierano: kwiecień 11, 2025, [https://en.wikipedia.org/wiki/User\\_interface\\_design](https://en.wikipedia.org/wiki/User_interface_design)
13. What is UI Design? | A Guide to User Interface Design - Maze, otwierano: kwiecień 11, 2025, <https://maze.co/collections/ux-ui-design/ui-design/>
14. What is User Interface (UI) Design? | IxDF - The Interaction Design Foundation, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/topics/ui-design>
15. What is user experience (UX) design? 2025 guide - Lyssna, otwierano: kwiecień 11, 2025, <https://www.lyssna.com/blog/what-is-user-experience-design/>
16. Don Norman: The term "UX" - YouTube, otwierano: kwiecień 11, 2025,

- <https://www.youtube.com/watch?v=9BdtGjoiN4E&pp=0gcJCdgAo7VqN5tD>
17. (PDF) FROM THE DEFINITION OF USER EXPERIENCE TO A FRAMEWORK TO CLASSIFY ITS APPLICATIONS IN DESIGN - ResearchGate, otwierano: kwiecień 11, 2025,  
[https://www.researchgate.net/publication/353687271\\_FROM\\_THE\\_DEFINITION\\_OF\\_USER\\_EXPERIENCE\\_TO\\_A\\_FRAMEWORK\\_TO\\_CLASSIFY\\_ITS\\_APPLICATIONS\\_IN\\_DESIGN](https://www.researchgate.net/publication/353687271_FROM_THE_DEFINITION_OF_USER_EXPERIENCE_TO_A_FRAMEWORK_TO_CLASSIFY_ITS_APPLICATIONS_IN_DESIGN)
  18. Where Did the Term “User Experience” Come From? - the Adobe Blog, otwierano: kwiecień 11, 2025,  
<https://blog.adobe.com/en/publish/2017/08/28/where-did-the-term-user-experience-come-from>
  19. www.interaction-design.org, otwierano: kwiecień 11, 2025,  
<https://www.interaction-design.org/literature/book/the-glossary-of-human-computer-interaction/user-experience-ux#:~:text=%22User%20Experience%22%2C%20often%20abbreviated,entry%20on%20User%20Experience%20%2F%20UX>
  20. User experience design - World Design Organization (WDO), otwierano: kwiecień 11, 2025, <https://wdo.org/glossary/user-experience-design/>
  21. Foundations of User Experience (UX) Design - Coursera, otwierano: kwiecień 11, 2025, <https://www.coursera.org/learn/foundations-user-experience-design>
  22. The Basics of User Experience Design - Riskcue, otwierano: kwiecień 11, 2025,  
<https://riskcue.id/uploads/ebook/20220201181424-2022-02-01ebook181200.pdf>
  23. User Experience - UX | The Glossary of Human Computer Interaction, otwierano: kwiecień 11, 2025,  
<https://www.interaction-design.org/literature/book/the-glossary-of-human-computer-interaction/user-experience-ux>
  24. The 6 key UX design principles to follow - Falmouth University, otwierano: kwiecień 11, 2025, <https://www.falmouth.ac.uk/news/6-key-ux-design-principles-follow>
  25. 7 fundamental user experience (UX) design principles all designers should know (2024), otwierano: kwiecień 11, 2025,  
<https://www.uxdesigninstitute.com/blog/ux-design-principles/>
  26. UX Design Principles | UXtweak, otwierano: kwiecień 11, 2025,  
<https://www.uxtweak.com/ux-design/principles/>
  27. Explore 10 Fundamental UX Design Principles for Optimal Use - Imaginovation, otwierano: kwiecień 11, 2025, <https://imaginovation.net/blog/ux-design-principles/>
  28. UX Design Principles for 2025 - UXPin, otwierano: kwiecień 11, 2025,  
<https://www.uxpin.com/studio/blog/ux-design-principles/>
  29. UX Design Principles: The 10 Rules Behind Products Users Love, otwierano: kwiecień 11, 2025, <https://userpilot.com/blog/ux-design-principles/>
  30. The Best Practices and Key Principles of UX Design - Baymard, otwierano: kwiecień 11, 2025, <https://baymard.com/learn/ux-design-principles>
  31. What core UX principles and theories do you always invoke in your Web Design work?, otwierano: kwiecień 11, 2025,  
[https://www.reddit.com/r/userexperience/comments/scm9ly/what\\_core\\_ux\\_principles\\_and\\_theories\\_do\\_you/](https://www.reddit.com/r/userexperience/comments/scm9ly/what_core_ux_principles_and_theories_do_you/)
  32. www.taazaa.com, otwierano: kwiecień 11, 2025,

<https://www.taazaa.com/ux-in-custom-software-development/#:~:text=During%20the%20development%20phase%2C%20UX,gathering%20user%20feedback%20for%20refinement>.

33. The Role of User Experience (UX) in Custom Software Development - Taazaa, otwierano: kwiecień 11, 2025, <https://www.taazaa.com/ux-in-custom-software-development/>
34. Software product development lifecycle — the UX perspective - CodiLime, otwierano: kwiecień 11, 2025, <https://codilime.com/blog/software-product-development-lifecycle-ux-perspective/>
35. The impact of UX and UI design on software development ..., otwierano: kwiecień 11, 2025, <https://devtalents.com/ux-and-ui-in-software-development/>
36. Software Development Lifecycle (SDLC) - Design Phase - Teaching Agile, otwierano: kwiecień 11, 2025, <https://teachingagile.com/sdlc/design>
37. UX/UI and Software Development: A Process Guide - Baytech Consulting, otwierano: kwiecień 11, 2025, <https://www.baytechconsulting.com/blog/a-guide-to-ux-ui-and-software-development-processes-and-practices-explained>
38. UX/UI Design Process & Deliverables in Software Development - MobiDev, otwierano: kwiecień 11, 2025, <https://mobidev.biz/blog/ui-ux-design-stages-deliverables-checklist>
39. UX design in the SDLC - Medium, otwierano: kwiecień 11, 2025, <https://medium.com/life-in-ux/ux-design-in-the-sdlc-life-in-47e5ab04d1a4>
40. Importance of UI/UX Design in Software Development - Carmatec, otwierano: kwiecień 11, 2025, <https://www.carmatec.com/blog/importance-of-ui-ux-design-in-software-development/>
41. What is SDLC? Understanding The Phases Of The Software Development Life Cycle, otwierano: kwiecień 11, 2025, <https://www.altigee.com/magazine/what-is-sdlc-understanding-the-phases-of-the-software-development-life-cycle>
42. In which phases of Software Development Life Cycle is UXD Involved [closed] - User Experience Stack Exchange, otwierano: kwiecień 11, 2025, <https://ux.stackexchange.com/questions/121524/in-which-phases-of-software-development-life-cycle-is-uxd-involved>
43. How User Research Elevates UX in Your Software Product | Iterators, otwierano: kwiecień 11, 2025, <https://www.iteratorshq.com/blog/mastering-user-research-improve-users-experience-in-of-your-software-product/>
44. User Research: What It Is and Why You Should Do It - The Interaction Design Foundation, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/article/user-research-what-it-is-and-why-you-should-do-it>
45. What is UX Research, Why it Matters, and Key Methods | Maze, otwierano: kwiecień 11, 2025, <https://maze.co/guides/ux-research/>

46. The Beginner's Guide to Capturing UX Requirements - UXPin, otwierano: kwiecień 11, 2025,  
<https://www.uxpin.com/studio/blog/the-beginners-guide-to-capturing-ux-requirements/>
47. Requirements gathering: a beginner's guide for UX Designers | by Kimber Capuchino | Bootcamp | Medium, otwierano: kwiecień 11, 2025,  
<http://medium.com/design-bootcamp/requirements-gathering-a-beginners-guide-for-ux-designers-48556f040da>
48. User Research Methods and Recommendations | Enterprise Architecture - Harvard University, otwierano: kwiecień 11, 2025,  
<https://enterprisearchitecture.harvard.edu/user-research-methods-and-recommendations>
49. Why user research is essential for product development - Optimal Workshop, otwierano: kwiecień 11, 2025,  
<https://www.optimalworkshop.com/blog/why-user-research-is-essential-for-product-development>
50. The role of user involvement in requirements quality and project success - ResearchGate, otwierano: kwiecień 11, 2025,  
[https://www.researchgate.net/publication/4186939\\_The\\_role\\_of\\_user\\_involvement\\_in\\_requirements\\_quality\\_and\\_project\\_success](https://www.researchgate.net/publication/4186939_The_role_of_user_involvement_in_requirements_quality_and_project_success)
51. The Impact of the User Experience (UX) on the Quality of the Requirements Elicitation, otwierano: kwiecień 11, 2025,  
[https://www.researchgate.net/publication/338955160\\_The\\_Impact\\_of\\_the\\_User\\_Experience\\_UX\\_on\\_the\\_Quality\\_of\\_the\\_Requirements\\_Elicitation](https://www.researchgate.net/publication/338955160_The_Impact_of_the_User_Experience_UX_on_the_Quality_of_the_Requirements_Elicitation)
52. User understanding of IS needs and expectations: impact on quality of requirements and satisfaction with system - Carolina Digital Repository, otwierano: kwiecień 11, 2025,  
[https://cdr.lib.unc.edu/concern/masters\\_papers/ng451n056?locale=en](https://cdr.lib.unc.edu/concern/masters_papers/ng451n056?locale=en)
53. Software For Real People: User Stories in UX Design - StoriesOnBoard, otwierano: kwiecień 11, 2025, <https://storiesonboard.com/blog/user-stories-in-ux-design>
54. User Stories | Examples and Template - Atlassian, otwierano: kwiecień 11, 2025,  
<https://www.atlassian.com/agile/project-management/user-stories>
55. The Ultimate Guide to UX User Stories [With Examples] - CareerFoundry, otwierano: kwiecień 11, 2025,  
<https://careerfoundry.com/en/blog/ux-design/ultimate-guide-ux-user-stories/>
56. What Is a User Story? Understanding Its Role in Agile Development and UX - Coursera, otwierano: kwiecień 11, 2025,  
<https://www.coursera.org/articles/what-is-user-story>
57. What Are User Stories? — updated 2025 | IxDF - The Interaction Design Foundation, otwierano: kwiecień 11, 2025,  
<https://www.interaction-design.org/literature/topics/user-stories>
58. Chapter 15: Requirements and user stories - Agile Business Consortium, otwierano: kwiecień 11, 2025,  
<https://www.agilebusiness.org/dsdm-project-framework/requirements-and-user-stories.html>

59. Understanding user stories in UX design - LogRocket Blog, otwierano: kwiecień 11, 2025, <https://blog.logrocket.com/ux-design/understanding-user-stories/>
60. Should design be included in a user story? - User Experience Stack Exchange, otwierano: kwiecień 11, 2025, <https://ux.stackexchange.com/questions/106044/should-design-be-included-in-a-user-story>
61. Who's typically responsible for writing user stories? What about intake for experience reviews or design briefs? : r/UXDesign - Reddit, otwierano: kwiecień 11, 2025, [https://www.reddit.com/r/UXDesign/comments/15fy0fb/whos\\_typically\\_responsible\\_for\\_writing\\_user/](https://www.reddit.com/r/UXDesign/comments/15fy0fb/whos_typically_responsible_for_writing_user/)
62. UX vs. UI Design - The Differences You Need to Know - UXPin, otwierano: kwiecień 11, 2025, <https://www.uxpin.com/studio/blog/ux-vs-ui-differences/>
63. UI vs. UX: What's the difference between user interface and user experience? - UserTesting, otwierano: kwiecień 11, 2025, <https://www.usertesting.com/blog/what-is-ui-vs-ux>
64. www.interaction-design.org, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/article/user-interface-designer#:~:text=User%20interface%20design%20%E2%80%94or%20UI,screen%20elements%20put%20together%20in>
65. What Does a User Interface (UI) Designer Do? | IxDF - The Interaction Design Foundation, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/article/user-interface-designer>
66. What Is UI Design? Definition, Tips, Best Practices - Coursera, otwierano: kwiecień 11, 2025, <https://www.coursera.org/articles/ui-design>
67. UX vs UI: What's the Difference? | IxDF - The Interaction Design Foundation, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/article/ux-vs-ui-what-s-the-difference>
68. Solved What is User Interface Design? According to the folks | Chegg.com, otwierano: kwiecień 11, 2025, <https://www.chegg.com/homework-help/questions-and-answers/user-interface-design-according-folks-interaction-design-foundation-wwwinteraction-designorg-q119217490>
69. User Interface - Product Glossary by Rapidr, otwierano: kwiecień 11, 2025, <https://rapidr.io/glossary/product/user-interface/>
70. What Is User Interface Design (Definition, Types)? | Built In, otwierano: kwiecień 11, 2025, <https://builtin.com/articles/user-interface-design>
71. User interface - Wikipedia, otwierano: kwiecień 11, 2025, [https://en.wikipedia.org/wiki/User\\_interface](https://en.wikipedia.org/wiki/User_interface)
72. A Quick Guide to UI Design Fundamentals | Blush Blog, otwierano: kwiecień 11, 2025, <https://blush.design/blog/post/guide-ui-design>
73. Importance of Good User Interface Design in Business Success - UX 4Sight, otwierano: kwiecień 11, 2025, <https://ux4sight.com/blog/designing-a-good-user-interface-and-why-its-vital-for>

[-business](#)

74. User Interface Design Guidelines: 10 Rules of Thumb | IxDF, otwierano: kwiecień 11, 2025,  
<https://www.interaction-design.org/literature/article/user-interface-design-guidelines-10-rules-of-thumb>
75. What are the Nielsen's 10 Usability Principles? | Aguayo Blog, otwierano: kwiecień 11, 2025,  
<https://aguayo.co/en/blog-aguayo-user-experience/what-are-the-10-usability-principles-by-nielsen/>
76. User interface design - Object-Oriented Design and Data Structures, otwierano: kwiecień 11, 2025, <https://andrewcmeyers.github.io/oodds/lecture.html?id=uidesign>
77. What Is User Interface (UI) Design? A Comprehensive Guide - CareerFoundry, otwierano: kwiecień 11, 2025,  
<https://careerfoundry.com/en/blog/ui-design/what-is-ui-design-guide/>
78. What is a User Interface Design? Key UI Principles - Userpilot, otwierano: kwiecień 11, 2025, <https://userpilot.com/blog/user-interface-design/>
79. User Interface Elements Every Designer Should Know - UXPin, otwierano: kwiecień 11, 2025,  
<https://www.uxpin.com/studio/blog/user-interface-elements-every-designer-should-know/>
80. 32 UI Elements Designers Need To Know: 2025 Guide - CareerFoundry, otwierano: kwiecień 11, 2025,  
<https://careerfoundry.com/en/blog/ui-design/ui-element-glossary/>
81. Mastering UI Design: A Comprehensive Guide to User Interface Elements | Clay, otwierano: kwiecień 11, 2025, <https://clay.global/blog/ux-guide/ui-elements>
82. What are UI elements? User interface elements explained - Uizard, otwierano: kwiecień 11, 2025,  
<https://uizard.io/blog/what-are-ui-elements-ui-design-elements-explained/>
83. 10 Interactive Design Examples to Inspire You - Userpilot, otwierano: kwiecień 11, 2025, <https://userpilot.com/blog/interactive-design-examples/>
84. Components — Material Design 3, otwierano: kwiecień 11, 2025,  
<https://m3.material.io/components>
85. Interactive UI Design, otwierano: kwiecień 11, 2025,  
<https://www.koombea.com/blog/interactive-ui-design/>
86. UX Design: Types of Interactive Content Amplifying Engagement - Tubik Blog, otwierano: kwiecień 11, 2025,  
<https://blog.tubikstudio.com/ux-design-interactive-content/>
87. Laws of UX: Home, otwierano: kwiecień 11, 2025, <https://lawsofux.com/>
88. Building Top Software Delivery Teams - BairesDev, otwierano: kwiecień 11, 2025,  
<https://www.bairesdev.com/blog/building-top-software-delivery-teams/>
89. What Is a Software Delivery Team? Roles, Benefits & Best Practices, otwierano: kwiecień 11, 2025,  
<https://www.jalasoft.com/blog/what-is-a-software-delivery-team>
90. The 8 Essential Roles of a Software Delivery Team - CTO Fraction, otwierano: kwiecień 11, 2025,

- <https://ctofraction.com/blog/the-8-essential-roles-of-a-software-delivery-team/>
91. Software Development Team Structure - Roles and Responsibilities - Enkonix, otwierano: kwiecień 11, 2025, <https://enkonix.com/blog/software-development-team-structure>
  92. Delivery Lead vs Product Manager Roles in Continuous Discovery and Delivery - Eleken, otwierano: kwiecień 11, 2025, <https://www.eleken.co/blog-posts/whats-the-difference-between-product-manager-and-delivery-lead-in-continuous-discovery-and-delivery-teams>
  93. Software Development Teams: Roles and Responsibilities - DevDynamics, otwierano: kwiecień 11, 2025, <https://devdynamics.ai/blog/software-development-teams-roles-responsibilities-and-characteristics/>
  94. 11 Key Roles in a Software Development Team - Alcor BPO, otwierano: kwiecień 11, 2025, <https://alcor-bpo.com/10-key-roles-in-a-software-development-team-who-is-responsible-for-what/>
  95. What Is a Delivery Team? The Key to Improving Product Delivery - Liatrío, otwierano: kwiecień 11, 2025, <https://www.liatrío.com/resources/blog/what-is-a-delivery-team-the-key-to-improving-product-delivery>
  96. Software Development Team Structure: Roles & Responsibilities - ITRex Group, otwierano: kwiecień 11, 2025, <https://itrexgroup.com/blog/software-development-team-structure/>
  97. Right and Responsibilities of a Delivery Team (Sandro Mancuso) - YouTube, otwierano: kwiecień 11, 2025, <https://www.youtube.com/watch?v=pX284uiCNqw>
  98. Key Roles and Responsibilities in a Software Development Team - Codewave, otwierano: kwiecień 11, 2025, <https://codewave.com/insights/software-development-team-roles-responsibilities/>
  99. Software Development Team Structure - Important Roles and ..., otwierano: kwiecień 11, 2025, <https://www.deazy.com/knowledge-hub/software-development-team-roles-and-responsibilities>
  100. Software Development Team Roles and Responsibilities - Revelo, otwierano: kwiecień 11, 2025, <https://www.revelo.com/blog/development-team>
  101. Software Development Team Roles, otwierano: kwiecień 11, 2025, <https://softwareengineering.stackexchange.com/questions/415792/software-development-team-roles>
  102. Development Team responsibilities | Scrum.org, otwierano: kwiecień 11, 2025, <https://www.scrum.org/forum/scrum-forum/6159/development-team-responsibilities>
  103. What roles are needed to make up a good software development team? - Reddit, otwierano: kwiecień 11, 2025, [https://www.reddit.com/r/softwaredevelopment/comments/ygnlnb/what\\_roles\\_are\\_needed\\_to\\_make\\_up\\_a\\_good\\_software/](https://www.reddit.com/r/softwaredevelopment/comments/ygnlnb/what_roles_are_needed_to_make_up_a_good_software/)

104. Top 10 UI/UX Services Companies In USA - TechAhead, otwierano: kwiecień 11, 2025, <https://www.techaheadcorp.com/top-10-ui-ux-service-companies/>
105. Top 12 Companies For Outsourcing UI/UX Development Services in ..., otwierano: kwiecień 11, 2025, <https://www.hirewithnear.com/blog/outsource-ui-ux-development-services>
106. What are UX Designers? — updated 2025 | IxDF - The Interaction Design Foundation, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/topics/ux-designers>
107. Agile & UX: a failed marriage?. This article analyses the integration... | by Aviram (Avi) Vijh, otwierano: kwiecień 11, 2025, <https://uxdesign.cc/agile-ux-a-failed-marriage-bd0db1182e5a>
108. Integrating UX Design into a Continuous-Delivery Environment - UXmatters, otwierano: kwiecień 11, 2025, <https://www.uxmatters.com/mt/archives/2019/06/integrating-ux-design-into-a-continuous-delivery-environment.php>
109. Dev & UX: How Integrating UX Improves Engineering's Efficiency and Sanity - InfoQ, otwierano: kwiecień 11, 2025, <https://www.infoq.com/articles/integrating-development-UX/>
110. The Role of DevOps in UX/UI Design - Ironhack, otwierano: kwiecień 11, 2025, <https://www.ironhack.com/us/blog/the-role-of-devops-in-ux-ui-design>
111. What are User Interface (UI) Design Patterns? | IxDF - The Interaction Design Foundation, otwierano: kwiecień 11, 2025, <https://www.interaction-design.org/literature/topics/ui-design-patterns>
112. Achieving Harmony: Balancing Business Goals and the User Experience in Design, otwierano: kwiecień 11, 2025, <https://www.uxmatters.com/mt/archives/2024/02/achieving-harmony-balancing-business-goals-and-the-user-experience-in-design.php>
113. How to integrate UI Automation into your CI/CD Pipeline for Continuous Testing, otwierano: kwiecień 11, 2025, <https://www.sherwen.com/insights/web-development/how-to-integrate-ui-automation-into-your-ci-cd-pipeline>
114. 10 CI/CD Pipeline Examples To Help You Get Started - Zeet.co, otwierano: kwiecień 11, 2025, <https://zeet.co/blog/ci-cd-pipeline-examples>
115. Scalable CI/CD Pipeline Examples: Improve the Development Churn - Simform, otwierano: kwiecień 11, 2025, <https://www.simform.com/blog/scalable-ci-cd-pipeline-examples/>
116. What are the Differences Between UX and UI? - Meadows School of ..., otwierano: kwiecień 11, 2025, <https://www.smu.edu/meadows/newsandevents/news/2025/what-are-the-differences-between-ux-and-ui>
117. UI vs UX Design: Differences, Similarities, and Synergies - Hotjar, otwierano: kwiecień 11, 2025, <https://www.hotjar.com/ui-design-vs-ux-design/>
118. UI vs. UX design | Differences and similarities - edX, otwierano: kwiecień 11, 2025, <https://www.edx.org/resources/ux-design-vs-ui-design>
119. UI vs. UX Design: What's the Difference? | Coursera, otwierano: kwiecień 11,

- 2025, <https://www.coursera.org/articles/ui-vs-ux-design>
120. UX vs UI Design: What's the Difference? - Maze, otwierano: kwiecień 11, 2025, <https://maze.co/blog/ui-vs-ux/>
121. UX vs UI: What's the Difference? - Designlab, otwierano: kwiecień 11, 2025, <https://designlab.com/blog/what-is-the-difference-between-ui-and-ux>
122. UX vs UI: Key Differences, Similarities, And Where They Overlap - The Product Manager, otwierano: kwiecień 11, 2025, <https://theproductmanager.com/topics/ux-vs-ui-key-differences-similarities-and-where-they-overlap/>
123. eli5 the difference between ui and ux : r/explainlikeimfive - Reddit, otwierano: kwiecień 11, 2025, [https://www.reddit.com/r/explainlikeimfive/comments/s5c9qq/eli5\\_the\\_difference\\_between\\_ui\\_and\\_ux/](https://www.reddit.com/r/explainlikeimfive/comments/s5c9qq/eli5_the_difference_between_ui_and_ux/)